Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/08/2005
PI Name:	Rutzky, Lynne P. Ph.D.		
Project Title:	Effect of Microgravity on Pancreatic Islet Xenotransplantation, Vascularization, and Stem Cell Growth		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Lynne.P.Rutzky@uth.tmc.edu	**Fax:**	FY 713-500-0784
PI Organization Type:	UNIVERSITY	**Phone:**	713-500-7398
Organization Name:	University of Texas Health Science Center at Houston		
PI Address 1:	6431 Fannin Street, Room 6.240 MSB		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Kirby, Mary	**Contact Phone:**	
Contact Email:	mary.f.kirby@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Katz, Stephen M. (The University of Texas Health Science Center at Houston) Stepkowski, Stanislaw M. (The University of Texas Health Science Center at Houston)		
Grant/Contract No.:	NAG0-1353		
Performance Goal No.:			
Performance Goal Text:			
	Introduction and Background: Intensive insulin replacement therapy only delays the onset of severe multiple organ complications among patients afflicted with Type I diabetes mellitus. Currently, reliable cure of diabetes requires whole organ pancreas allotransplantation despite the fact that the insulin-producing pancreatic islets comprise less than 5% of all pancreatic tissue. Although over the last five years there has been significant progress in successful transplantation of allogeneic under the guidance of the Edmonton Protocol, which is steroid-free and low in the calcineuin-inhibitor tacrolimus. Unfortunately, there is a chronic shortage of pancreatic islet with a limited number of human donors available. Development of xenotransplantation protocols with pig islets and the production of healthy, more potent islets for clinical allotransplant ion are the focus of much research effort. In contrast, auto-transplantation of islets following total pancreatectomy in-patients with acute pancreatitis has been successful, achieving an insulin replacement-free		

Task Description:

status. These results demonstrate that non-immunogenic, syngeneic islets in a non-autoimmune environment that are purified by current techniques of collagenase digestion and automated pancreatic islet processing survive in vivo and can reverse diabetes. Therefore, a major obstacle to clinical islet transplantation is immunogenicity of allogeneic islets. While pre-transplant culture improves survival of islets by reducing immunogenicity, with standard stationary islet culture there is progressive loss of islets and function. These problems may be avoided by use of a rotating wall vessel (RWV) modeled microgravity bioreactor developed by NASA to culture cells while in continuous free fall. High mass transfer of nutrients characterizes bioreactor culture with significantly reduced turbulence, permitting cells to interact and form 3-dimensional (3-D) tissue-like configurations and in some instances to differentiate.
Our recent work (Transplantation, 74:13-21, 2002) shows clearly that mouse islets cultured in bioreactors have long-term allograft survival and well preserved ultrastructural morphology and function, similar to fresh islets. Moreover, bioreactor-cultured islets develop intercellular channels with microvilli, which open onto the islet surface and may function to improve nutrition and facilitate graft vascularization. In contrast, stationary dish-cultured islets are less well preserved with signs of ultrastructural degeneration, contain less endocrine hormones, and they do not form large intercellular channels. To overcome chronic shortages of islets for clinical transplant, xenotransplantation of pig islets into Type 1 diabetics and proliferation and expansion of cultured human pancreatic adult stem cells into insulin producing cell arrays are two promising approaches to address chronic shortages of donor islets.

The present research investigates several avenues of islet xenotransplantation, including rat: mouse, human: mouse, and pig: mouse models, namely: 1) to reduce islet immunogenicity and improve preservation of morphology and function by culturing donor islets in microgravity bioreactors prior to transplantation, 2) to co-culture islets in bioreactors with recipient mouse microvascular or vascular endothelial cells and to seed large channels formed during bioreactor culture in islet grafts with vascular endothelium to facilitate engraftment. This research project also uses microgravity bioreactors to study adult ductal stem cell transdifferentiation into arrays of insulin-producing cells isolated from rat, pig, or human non-islet pancreas acinar tissues, as recently reported in monolayer culture. The goals of this research project are threefold: 1) to evaluate the effect of modeled microgravity culture on pancreatic islets and to assess whether this culture system reduces immunogenicity, preserves morphology and function and produces an improved source of islets for xenotransplantation; 2) to study the interaction of vascular endothelial cells and microvascular endothelial cells with pancreatic islets cultured in RWV bioreactors and to determine the effect of co-culture of islets with endothelial cells on graft vascularization and survival; and 3) to investigate the potential of microgravity bioreactors to facilitate transdifferentiation of adult pancreatic ductal stem cells into insulin-producing cell arrays for transplantation. We have shown previously that modeled microgravity culture in RWV bioreactors eliminates dendritic antigen presenting cells and Class II major histocompatibility antigen bearing cells and reduces islet immunogenicity in a mouse allograft transplant model system. In addition, islets cultured for 7 days in bioreactor had superior morphology, endocrine hormone content, and function. Ultrastructural examination of bioreactor or dish cultured islets and fresh islets showed that only bioreactor islets exhibited channels between arrays of 3-4 adjacent islet cells, which may be related to improved islet nutrition and function. Since islets are in continuous free fall at terminal settling velocity in bioreactor culture, the channels may permit culture medium to continuously flow through islets, maximizing gas and nutrient exchange.

Task Significance: The hypothesis of this proposal is that microgravity bioreactor-culture of rat and pig islets in RWV bioreactors: 1) reduces islet immunogenicity, while preserving morphology and function, making them an excellent donor tissue source for long-term islet xenotransplantation; 2) forms large, sinusoidal channels within islets which act as a scaffold for co-cultured recipient endothelial cells, facilitating vascularization and engraftment; and 3) promotes proliferation and differentiation of ductal stem cells into insulin-producing cell arrays. The significance of this research is that microgravity bioreactor culture may provide a superior source of donor islets for xenotransplantation by reducing immunogenicity, improving islet physiology, and providing a network of cell intra-islet channels for endothelial cell seeding and enhanced vascularization. In addition this research project may help to expand the number of insulin secreting cells available for transplant by providing an environment for transdifferentiation of adult ductal cells derived from acinar tissues produced during islet isolation into insulin secreting cell arrays. The overall objective of this proposal is to exploit the unique biological effects of modeled microgravity (MMG) culture in rotating wall vessel (RWV) bioreactors to produce: 1) physiologically superior pancreatic islets with reduced immunogenicity, improved vascularization, and faster engraftment for xenotransplantation and 2) to stimulate islet stem cells to transdifferentiate into insulin-producing cell arrays.

Critical Path Roadmap: Space flight, RWV bioreactor-culture and hypokinesia, bed rest modeled-microgravity are associated with alterations in immune function, endocrine systems, and plasma glucose and insulin levels. Simulated microgravity culture and space flight are associated with reversible glucose metabolic alterations and hyperglycemia. When young rats were flow in the space shuttle for 16 days, several organs, including pancreas weighed less than ground controls, indicating changes in organ growth patterns. The Critical Path Roadmap (CPR) does not specifically address the previously described in-flight space and ground-based phenomena associated with altered insulin and glucose metabolism associated with hyperglycemia and/or a subclinical prediabetic state. We believe that this potentially serious health-related issue should be included in the Critical Questions being developed to identify and ameliorate hazards associated with extended space travel to Mars. It is possible that muscle mass loss and inactivity, hormonal imbalance, nutritional deficiencies, and reduced immune surveillance, delayed wound healing and microvascular damage may be further exacerbated by space flight-induced hyperglycemia and/or subclinical diabetes. Therefore, we propose that CPR questions may be answered by studying ground-based models comparing function of normal rodent islets with islets from bioreactor cultures and from rodents housed in tail suspension modeled microgravity weightlessness conditions. Diabetic, syngeneic recipients could be used as a sensitive surrogate to measure islet function in diabetic animals in modeled tail suspension modeled microgravity conditions. These models could also be used to study alterations in immune surveillance, by monitoring possible prolonged islet allograft survival as a function of decreased immune surveillance. The pancreatic islet models developed during our studies could provide mechanistic information about glucose metabolism aberrations encountered during space flight by transplantation of islets to diabetic, syngeneic recipients tail suspended rodents. Incorporation of nutritional countermeasures with diets that include nucleotide supplements may decrease development of hyperglycemia and provide insights into the mechanism of this prediabetic state. It may be also be possible to answer questions associated with space flight-associated hyperglycemia by transplanting islets from transgenic mice that over-express insulin and/or glucagon to study glucose regulation and homeostasis in a tail-suspension or bioreactor model system.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	Benefits to life on Earth We have shown previously that RWV bioreactor culture of mouse pancreatic islets for 7 days provides islets for allograft transplantation with decreased immunogenicity and improved morphology and function, as compared with stationary dish culture. The current research builds on these observations, postulating that bioreactor culture provides improved quality islets for xenotransplantation. In addition, we are using the bioreactor system to study transdifferentiation of ductal cells found in non-islet pancreatic ascinar tissues into insulin secreting cell arrays to help expand the amount of insulin secreting tissues obtained from a donor pancreas. We are also investigating the utility of the bioreactor system to facilitate early vascularization by co-culturing islets with vascular endothelial cells. If successful, these approaches would help expand the pool of pancreatic islets or insulin secreting cell arrays and would be an important additional tissue resource for those individuals with Type I diabetes and who would benefit from islet allo- or xenotransplantation.
Task Progress:	Materials and Methods: In the current studies, recipient diabetic mice were transplanted under the kidney capsule with 500 fresh, dish- or bioreactor-cultured rodent or human islets, as described by us previously. Islets were cultured in 10 or 50 mL disposable HARV RWV bioreactors or a 35 mL STLV perfusion bioreactors for various intervals up to 7 days and used directly in allo- or xenotransplantation and histological studies. Islets were evaluated by dithizone staining to visualize insulin content and by immunostaining for insulin, glucagon, and somatostatin. Following transplant, islet function was followed by monitoring serial blood glucose levels, weight and by in vivo glucose tolerance tests. In some experiments islets were evaluated by light and electron microscopy for apoptosis. In some studies, the presence of endothelial cells within islets following co-culture was examined by immunostaining with endothelial cell marker antibodies, PECAM (CD31) and Factor VIII or lectin staining. In co-culture studies for xenotransplantation, rat islets were cultured in disposable bioreactors, co-cultured with syngeneic recipient aorta or mouse lung endothelial (MLEC) cells, and xenotransplanted under the kidney capsule of syngeneic diabetic Balb/c mice. Blood glucose levels and weight were followed at intervals to assess the status of the graft. Islets were evaluated histologically at the termination of the experiment for presence of endothelial cells and vascularization. Results and Discussion: During the past year, we continued to investigate the impact of bioreactor culture on apoptosis (programmed cell death) in rodent and human islets. While islet cell transplantation with the newer steroid-free, reduced immunosuppression regimens offers great clinical promise in treating type 1 diabetes, a chronic shortage of islets for clinical transplantation is exacerbated by damage by apoptosis, which probably accounts for considerable loss of functional islet mass during procurement, isolation and stationary flask culture, and engraftment. As we reported previously, culture of islets in disposable bioreactor improved rodent islet ultrastructure and hormone content and induced formation of nutritional channels. During the past year, apoptosis was evaluated in mouse and human islets by whole mount TUNEL staining for damaged DNA and evaluated by light microscopy and by TUNEL staining for electron microscopy. Bioreactor-cultured islets generally showed reduced the amount of apoptosis, as compared with stationary cultures. Moreover, in TUNEL-stained islet aggregates apoptotic cells were often located near the periphery and between adjacent islets, which was confirmed by ultrastructural analysis. In current studies, human or rodent islets were placed in stationary Petri dishes or disposable bioreactor culture. In some studies a reusable 35 mL STLV perfusion bioreactor system with a 500 mL culture medium reservoir (Synthecon, Houston, TX) was compared with the HARV disposable bioreactor and dish-cultured islets. When compared with stationary dish cultures, the perfusion bioreactor culture system reduced caspase activity, a central enzyme in the apoptosis cascade, and enhanced ATP levels in human islets during 7 days of culture. We believe that formation nutritional channels in bioreactor cultures increases penetration of nutrients and oxygen into the islet central core, preserving islet structure and function and reducing apoptosis. The density of human islets cultured in perfusion bioreactors was increased to a concentration estimated to be in the density range necessary to culture human islets isolated from single pancreas (~300,000 islet equivalents) in a theoretical perfusion bioreactor prior to clinical transplantation. We observed that when human islets were cultured at higher densities in the perfusion system, the islets continued to have reduced incidence of apoptosis and had smaller islet aggregates. In addition, they developed large nutritional channels between adjacent cells, similar to those observed earlier in mouse and human islet disposable HARV bioreactor-cultures at lower islet densities. These studies strongly suggest that it is feasible to scale up the perfusion system to accommodate and culture islets from a single pancreas for incubation prior to clinical transplantation. Additional studies to further reduce islet aggregation were conducted using various culture media to help reduce the tendency of islets to aggregate in bioreactor culture. These results indicate that culture of human islets in the bioreactor system will be beneficial during the immediate post-isolation period by improving islet quality prior to transplantation. During this time, it would be possible to prepare the recipient with various pre-transplantation immunosuppressive or tolerogenic protocols. The utility of bioreactor culture to facilitate vascularization of islets prior to transplantation to enhance early engraftment has been investigated during the past year. Previously, composite-tissue islets (human islets plus HUVEC human endothelial cells) were co-cultured and immunostained with two endothelial cell markers, Factor VIII and CD31 (PECAM) antibodies, to identify the presence of endothelial cells within islets. Our data suggest that bioreactor culturing may facilitate entrance and growth of HUVEC endothelial-cells via the surface pores and interior channels, providing preliminary evidence that the bioreactor system may be useful to seed islets with recipient endothelial cells for enhanced engraftment. During the past year, composite tissues of rat islets and mouse endothelial cells were produced by culturing rat islets in HARV disposable bioreactors, followed by co-culturing in the bioreactor with either mouse aorta endothelial cells or mouse lung endothelial (MLEC) cells. Composite rat:mouse islets were then xenotransplanted under the kidney capsule of diabetic syngeneic mouse recipients and blood sugar monitored at intervals to assess the functional status of the graft. No immunosuppression was used in these preliminary experiments. Composite tissues of rat:MLEC islets had prolonged graft survival when compared with fresh rat islets, dish-cultured or bioreactor-cultured rat islets. Histologic sections of rat:mouse islet composites are being evaluated by immunostaining and lectin staining. We have continued to investigate immune regulation occurring in the 7 day bioreactor-cultured mouse islet allograft model system with long-term allograft survival (mean survival time, MST>100 days) and without any exogenous immunosuppression. In the current studies, C57Bl/10 (H2b) islets were cultured in disposable RWV bioreactors (bioreactor) for 3 and 7 days prior to transplantation (500 islets under the kidney capsule) in diabetic Balb/c mice (H2d), a major histocompatibility mismatch strain combination. As we have shown in the past, freshly isolated mouse islet allografts transplanted into diabetic Stat6-/- or Stat4-/- (Signal Transducers and Activators of Transcription) gene knockout recipients were acutely rejected at about 12 days. Some Stat6-/- recipients of 3-day cultured islets showed delayed rejection, while some had long term allograft survival (>100 day). In contrast, all Stat4-/- recipients permanently accepted 3-day cultured islet allografts with long term survival, documenting involvement of Stat6-dependent regulation.

	When pure T cells from Stat4-/- tolerant recipients were adoptively transferred to diabetic SCID Balb/c recipients of fresh islets, graft survival was prolonged, as compared to SCID mice transferred with normal T cells. In addition, we also transplanted fully immunogenic fresh islets to Stat4-/- recipients treated with anti-CD40L monoclonal antibody (days 0, 2, and 4) and CTLA Ig (day 2), which block activation of the immune pathway Signal 2, producing immune tolerance. All of these recipients accepted islet allografts long term; whereas pure T cells from these drug-treated tolerant recipients adoptively transferred tolerance to SCID recipients of C57Bl10 islet allografts long term. We conclude that although spontaneous long-term survival is facilitated by depletion of donor dendritic cells in bioreactor culture, the potency of Stat6-driven T regulatory cells is increased by the presence of donor dendritic cells and treatment with tolerogenic protocols. Current studies are exploring the possibility of improved generation of regulatory T cells by the addition of immature dendritic cells to bioreactor-cultured islets along with tolerogenic protocols.
Bibliography Type:	Description: (Last Updated: 06/08/2005)
Abstracts for Journals and Proceedings	Rutzky, L.P.; Zhang, H.; Phan,T.; Kloc, M.; Stepkowski, S. "Dendritic Cells Facililitate Introduction of STAT6-Driven Regulation of Tolerance to Islet Allografts " American Journal of Transplantation 5 (supplement 11):313. 2005. , Jan-2005
Presentation	Rutzky, L.; Bilinski, S.; Kloc, M.; Zhang, H.; Phan,T.; Katz, S.; Stepkowski, S. "Modeled Microgravity Culture of Human Pancreatic Islets Improves Function and Reduces Apoptosis" 2005 NASA Biotechnology Cell Science Program Investigators Working Group, Galveston, TX, February 23-25. Feb-2005
Presentation	Rutzky,L.; Zhang, H.; Phan, T.; Kloc,M.; Stepkowski, S. "Dendritic Cells Facilitate Introduction of STAT6-Driven Regulation of Tolerance to Islet Allografts" American Transplant Congress, Seattle, WA, 21-25 May 2005 May-2005